AI-Assisted On-Call Queue: Budgeted Incident Detection for Cloud Microservices

(Alert Fatigue Reduction in Cloud Microservice Operations)

Final report

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Numbers come from scripts under `app/` and are checked by `python run_validate.py` against the *Checked numbers* table at the end. Part A follows the thirteen sections of the course brief; Part B holds the measured results; Part C says how to reproduce them.

Rubric criterion (weight)	Where it is answered
Problem definition and significance (10%)	§2; evidence in Part B §B6
Requirements, users and use cases (10%)	§3, §4; `docs/requirements.md` (20 FR, 11 NFR, 8 user stories, traceability to tests)
AI justification and technical approach (10%)	§5, §6
High-level production design (15%)	§6; `docs/architecture.md` (5 diagrams), `docs/adr/` (7 decisions), `docs/threat_model.md`
Deployment and scalability (10%)	§7; `docs/deployment.md`
Testing strategy (10%)	§8; `docs/test_plan.md`, `.github/workflows/ci.yml`
MLOps and lifecycle (20%)	§9; `docs/mlops.md`
Monitoring and evaluation (10%)	§10; `docs/monitoring.md`
Privacy and responsible AI (5%)	§11; `docs/privacy_responsible_ai.md`

Part A — The system

1. Project title and team

AI-Assisted On-Call Queue: Budgeted Incident Detection for Cloud Microservices. Solo project by Aishah Mohamed Alyammahi.

2. Problem definition

What the problem is. A cloud platform made of many small services raises thousands of alerts a day, but the engineer on call can read only a few. Real faults are missed because attention runs out, not because monitoring was silent.

Who experiences it. On-call engineers and small operations teams (primary), and the users of the platform when a fault is found late.

Why it matters. A fault that is noticed late lasts longer. Engineers who are paged for noise stop trusting the pager and start ignoring it ("alert fatigue").

How it is handled today. Alerts are escalated by log severity, and exact duplicates are deleted (Alertmanager grouping and routing, PagerDuty rules). Metric alerts are fixed thresholds or rate rules, such as `delta(memory[5m]) > 0.5 GB`.

Limits of the current approach. Severity rules read only logs. On the public GAIA microservice benchmark, **82% of injected faults never write a log line**; they show up only as a jump in a metric such as memory, so a log-severity rule catches **0%** of them at 20 items a day (§B1). Fixed metric thresholds either miss faults or flood the queue, and none of these rules knows that a person can read only K items a day.

How often. GAIA injects about 22 faults a day on ten services, more than a 20-item budget on 34 of 55 days (§B1).

If nothing changes. Teams keep choosing between missed faults and ignored pagers. More monitoring makes it worse, because every new signal competes for the same attention.

Why this is a software engineering project. The answer is not one model. It is a system: ingest alerts and metrics from existing tools, decide under a budget in real time, page safely, learn from engineers' feedback, and be deployable, testable, monitored and reversible.

> **Problem statement.** On-call engineers can read only a few alerts a day, but the rules that choose those alerts read log severity, which most real faults never trigger, so faults are missed while noise still reaches the pager.

3. Objectives and expected outcomes

Main objective. Design, build and evaluate an AI-assisted on-call queue that shows each engineer at most K items a day, in real time, and catches more real faults with fewer false items than the rules teams already run.

Specific objectives (measurable).

#	Objective	Target	Result
O1	Beat the rules a team already runs at a realistic budget	More faults caught at K = 20 than log severity, a static threshold and a Prometheus rate rule, with fewer false items	Detector 72.6% vs 68.9% (rate rule), 1.0 vs 2.1 false items a day; wins 8/8 walk-forward weeks (§B1, §B2)
O2	Decide in real time	No decision uses later data; confirmation within about a minute	Median 0.72 min; `tests/test_causality.py` (§B1)
O3	Engineer it for production	Authentication, safe paging, versioned models with rollback, monitoring endpoints, Docker	Done (§7, §9, §10)
O4	Test it properly	≥ 90% of tests passing, coverage ≥ 85%, p95 ≤ 1 s at 50 users	273/273, 90%, 0.32 s (§8)
O5	Let another team check it on its own data before trusting it	Evaluate my data works end to end without code, against a chance line	Done; NAB, SMD and RCAEval results reported, including failures (§B5)

Expected final product. A working web application and REST API (FastAPI) with a queue page, Connectors page and *Evaluate my data* page; a backend scoring service; an AI component (a cost-sensitive alert classifier with a model registry) next to rule-based metric detectors; a SQLite store; connectors for Alertmanager, Prometheus, PagerDuty, Slack and Opsgenie; a command-line tool; 273 automated tests and CI; a Docker deployment with per-environment settings; `/metrics` and `/monitoring`; technical documentation (`docs/`); and this report.

4. Target users and use case

Primary users. On-call engineers (site reliability or operations) on a team running microservices.

Secondary users. The team lead or operator, who installs the tool, decides when it may page and which model is active; platform owners who read the monitoring page; an evaluating team trying it on past data; examiners and researchers reproducing the results.

User	Needs
On-call engineer	A short list, in order, that contains the real faults; a reason for each item; a way to say "this was noise"
Operator / team lead	Safe installation; nothing paged by accident; secrets kept safe; a way to undo a bad model; health at a glance
Evaluating team	Proof on its own data, with a fair baseline and a chance line, before trusting it

System usage. Alertmanager posts alerts to the app all day; the app pulls memory metrics from Prometheus; a scoring job runs on each webhook batch (or on demand). The engineer opens the queue page during the shift and receives pages in Live mode. They provide nothing but marks: Real or Noise on items. The system produces the day's queue (metric jumps, slow leaks, monitoring gaps and escalated or "ask a person" alerts, at most K), each with a reason. After that, the engineer investigates the item; the mark becomes a label for the next model.

Use-case flow.

```
flowchart TD
    A[Alertmanager posts firing alerts] --> B[App checks the token and rejects bad rows]
    B --> C[Alerts buffered, duplicates skipped]
    C --> D[Scoring job pulls memory metrics from Prometheus]
    D --> E[Rules: metric jumps, slow leaks, monitoring gaps]
    D --> F[AI: alert classifier scores log alerts]
    E --> G[Causal queue: at most K a day, first come]
    F --> G
    G --> H{Live mode, channel and password?}
    H -- yes --> I[Page PagerDuty / Slack / Opsgenie with new items]
    H -- no --> J[Record what would have been paged]
    I --> K[Engineer reads the queue and marks Real / Noise]
    J --> K
    K --> L[Marks become labels; next model is trained, gated and versioned]
    L --> M[Job, decisions and metrics recorded for monitoring]
```

5. Why use AI?

What needs AI. Deciding which *log alerts* are worth a slot. Their text, source, severity and timing interact in ways fixed rules do not capture, and the right answer differs per team. A classifier trained on the team's own labels (and later its own Real / Noise marks) learns that, with a miss weighted ten times a false alarm.

What traditional software handles. Everything else, deliberately: ingesting webhooks, the metric detectors (a rise above the recent median is a rule, chosen on week 1 only), monitoring-gap detection, the daily budget, paging, storage and security. On GAIA the rules do most of the work: where metric data exists, the jump rule overlaps 99.6% of faults (§B6).

Why a rule-based approach alone is insufficient for alerts. Severity is noisy: "high" alerts are often benign and "info" lines appear during real faults. A fixed severity rule cannot use message text or the fact that an alert arrived during a memory jump, and it cannot adapt when engineers mark items as noise.

AI technology. Supervised classification: logistic regression on TF-IDF text features, source, severity, recent alert counts and an inside-a-jump flag (scikit-learn). A second learned piece, the confirmation threshold, is a cost-minimising cut on the training window. An experimental jump ranker (logistic regression) was tried and gave no gain (§B4).

Existing model or API. An existing, well-understood model family (scikit-learn logistic regression) trained locally; no external AI API or LLM, so no data leaves the server and there is no per-call cost.

What the classifier adds, measured. An ablation on GAIA (`run_classifier_ablation.py`, §B4) keeps the whole queue and changes only how log alerts are chosen for their quarter of the slots. At K = 20 the queue with the classifier catches 69.6% with 1.8 false items a day; with log severity instead it catches 71.0% (1.4 false a day); with no log alerts at all, 71.2% (1.2 false a day). **On GAIA the classifier adds no catches; it costs about 1.6 points**

and 0.6 false items a day, because 82% of GAIA faults write no log line and the alerts it escalates spend slots that jumps would have used. Its value is therefore not shown on this benchmark. It is kept for teams whose faults *do* log: it is the only part that learns from that team's Real / Noise marks (a Noise mark becomes a negative label at the next retrain), and alerts can never take more than a quarter of the budget. **It is not on by default.** A trained model is used only if, on the held-out part of its batch, it beats the same queue *without* log alerts (§9). The first model has to pass too. On GAIA it does not (848 vs 872 of 1,223 held-out faults caught, §B4), so the tool keeps it as a candidate and scores GAIA with rules only.

Limitations of AI here. It needs labels of both classes, so a new team starts in rules-only mode; learned thresholds do not transfer between datasets (one learned on GAIA's 1 GB faults rejects 0.5 GB faults, §B3); and 82% of GAIA faults leave no log, which caps a log-only model near 18% (§B6).

6. AI component and technical approach

Question	Answer
AI technology	Cost-sensitive logistic regression for log alerts; rule-based metric detectors beside it
Existing model or own?	An existing model family, trained on the team's own labelled data; not a new algorithm
Input	Each alert: time, source, severity, type, message; whether it fell inside a confirmed memory jump; recent counts from the same source
Output	P(alert is a real fault), turned into an action: escalate, ask a person, or hide, with a reason
Data needed	Past alerts with incident windows, or Real / Noise marks; memory series for the detectors
How the software talks to it	In-process: `pipeline.score_alerts` calls the classifier; models are loaded from the registry (`product/model_store.py`)
If it is wrong	The queue is capped at K; uncertain alerts go to a person; engineers mark Noise; a model is not promoted unless it beats the queue without log alerts (and the active model) on held-out data, and the operator can roll back
Main limitation	Label-hungry; helps only services that log

How the queue decides (details in §B1 and `design_specification.md`). One queue, at most K items a day, first come, first served:

1. **Metric jumps.** A point is hot when it is more than **0.5 GB** above the median of the previous **15 minutes** on the same service. Two hot points no more than two minutes apart **confirm** a jump. A jump is shown at its confirmation point, and only if its rise at that point is at least a threshold learned on the training window (0.62 GB on week 1).
2. **Slow leaks.** The same rule with a **3-hour** median and 10 minutes of hot points catches memory that climbs over hours. A leak is dropped only when a fast jump on the same series was already shown when the leak was confirmed.
3. **Monitoring gaps.** A service with no metric point for **10 minutes** is shown as a gap item ("we are blind here"), also while the feed is still dead.
4. **Log alerts** may use at most **a quarter** of the slots (5 of 20), first come. An alert on a service whose shown jump is still open rides on that jump and uses no slot.

Nothing uses a later point, the end of a jump, or its peak (ADR 0001). For metrics that are not bytes, a relative mode (a share of the median) and a unit-free robust mode exist.

Training on GAIA. GAIA MicroSS (CloudWise), July–August 2021: 34,863 log lines, 1,367 injected faults, each holding **1 GB of extra memory for 600 s** on one service. Train on the first 7 days; test on the next 55 days (**1,223 faults**); days are never shuffled.

What is learned	How
Jump size 0.5 GB	Chosen on week 1 only (`run_memory.py --sweep`)
Jump confirmation threshold	Rise at confirmation with the lowest week-1 cost → 0.62 GB
Alert classifier and its threshold	Logistic regression on week-1 alerts; threshold with the lowest week-1 cost (a miss = 10 false alarms)
Static-rule baseline	Week-1 99th percentile per service

Production design. Two layers: a library that knows nothing about HTTP (`app/src/`) and a product layer (`app/product/`) with the web app (`product/web/`: security middleware, three route modules), the job pipeline (inputs, labels, paging), storage, connectors, the model registry and monitoring. Diagrams (context, components, ingest-to-page sequence, deployment, data model): `docs/architecture.md`. Decisions: `docs/adr/`. Threats and mitigations: `docs/threat_model.md`.

7. Deployment strategy

One Docker container per team, on the team's own server or VM, behind a TLS reverse proxy; SQLite and the model registry on a Docker volume; the model runs in-process; alerts arrive from Alertmanager and metrics are pulled from Prometheus; pages go to PagerDuty, Slack or Opsgenie. The image is built from a hash-pinned lock file.

	Development	Test	Production
Where	Laptop	GitHub Actions; a shadow instance next to production	Team server
Settings	`app/deploy/dev.env`	`app/deploy/test.env`	`app/deploy/prod.env` (from the example; not in git)
Paging	Never	Never (records *would have paged*)	Only after a shadow period and a deliberate Live switch

Measured capacity: 50 concurrent users at 228 requests/s with p95 0.32 s; a 400-alert webhook batch scores in 0.2 s and the full GAIA month in 27 s. The design runs one worker per team; the scaling path is one container per team, or PostgreSQL with shared locks for a shared deployment (ADR 0003).

AI governance. The operator is responsible for the AI. It may only order and select items in the queue (escalate, ask a person, hide with a reason); it never acts on infrastructure. Humans review everything in shadow mode, answer "ask a person" items, and mark Real / Noise. Every decision records its reason, score and the model version. New models pass a promotion gate and can be rolled back; the change is audited. No data is sent to an external AI service. Full answers: `docs/deployment.md`.

8. Testing strategy

Level	What	Examples
Unit	Detectors, queue, policy, scoring, settings	first come beats biggest first; a jump is confirmed at its second point
Integration	Frontend routes ↔ backend ↔ database, with outbound calls mocked	webhook → buffer → job → queue; Live fans out to every channel; model rollback through the API
System	The whole pipeline on GAIA, NAB, SMD and RCAEval;	walk-forward over 8 weeks; later data changes nothing

Level	What	Examples
	`experiment.py` on a synthetic run	
AI-specific	Wrong or poor outputs, unexpected input, edge cases	a malformed webhook row is rejected instead of blocking every later job; one-class labels fall back to rules; a worse model is not activated; hidden alerts stay visible
Security	Login, SSRF, CSRF, DNS rebinding, secrets, body limits	loopback tricks refused; forwarded requests need a password
Performance	Latency, concurrent users, throughput	`perf/load_test.py`: 20 and 50 users

Hallucination and prompt injection do not apply (no generative model), but the equivalent risks are tested: untrusted webhook content cannot break or steer the queue, and wrong model outputs are bounded by the budget and the promotion gate.

Criterion	Target	Result
Tests passing	100%	273 / 273 (Python 3.10 and 3.12)
Line coverage	≥ 85%	90%
Type errors	0	0
p95 latency at 50 users	≤ 1 s	0.32 s
Error rate under load	≤ 1%	0%
Report numbers match results	every checked number	93 / 93

CI runs lint, type checks, the tests on two Python versions with coverage, `run_validate.py`, a dependency audit (`pip-audit` on the hash-pinned lock) and a Docker build with a health check on every push. The audit step was added on 7 October with an option `pip-audit` does not have, so that job failed on that commit; it was corrected on 10 October (a hand run of the corrected command found no known vulnerabilities). Details: `docs/test_plan.md`.

9. MLOps

- **Versioning.** Code in Git through pull requests; dependencies in `uv.lock` and a hash-pinned `requirements.lock`; configuration in `src/config.py` and per-environment files; data by fingerprint in the experiment log; **models as numbered versions with model cards** in `models/registry.json` (training time, data fingerprint, code version, held-out results, gate decision). No prompts (no LLM).
- **Reproducibility.** `uv sync --group dev`, `uv run pytest`, `uv run python run_validate.py`, `docker compose up`; every environment variable is documented; the public datasets are fetched by scripts (`tools/fetch_gaia.py`, `fetch_nab.py`, `fetch_smd.py`, `fetch_rcaeval_mem.py`), and `fetch_gaia.py` checks the SHA-256 of every GAIA file used.
- **Model updates.** A new model is trained when a batch has labels of both classes, on the batch's first 7 days, and tested on the rest. It becomes active, and scores the batch, only if on that held-out window (1) it costs less than showing no log alerts (a miss = 10, a false alarm = 1); (2) the whole queue with its alerts costs less than the queue without log alerts (a missed fault = 10, a false item = 1); and (3) it costs no more than the active model on the same held-out alerts. The first model is not promoted just because none is active. A model that fails stays a candidate, and the batch is scored by the active model, or by rules only. On GAIA the classifier passes (1) and fails (2) (§B4), so rules only stays. The operator can roll back to any version (`POST /models/{version}/activate`, hash-checked, audited). Precision from engineers' marks and alert-volume drift are monitored, and a retrain is suggested after 7 days or new Real marks.
- **Testing before release.** CI for code; the promotion gate and shadow mode for models.

- **Experiment tracking.** Every experiment script appends to `app/results/experiments.jsonl`: commit, library versions, configuration, data fingerprint, output hashes, headline metrics and the reason for the run.

Details: `docs/mlops.md`.

10. Monitoring and evaluation

`GET /metrics` (Prometheus) and `GET /monitoring` (checked summary) cover:

Area	Signals	Target
AI performance	Precision from Real / Noise marks; model used and version per job; promotion decisions	≥ 0.70 with ≥ 20 marks
Software / system	Latency by route, 5xx rate, job duration and outcome, pages sent and failed, `/health`	p95 ≤ 1 s; 5xx ≤ 2%; 0 failed pages
Data	Alert-volume drift (last batch vs median), monitoring gaps, rejected and duplicate rows, buffer size	ratio 0.5–2; ≤ 3 gaps
AI / model	Output mix per job (escalated / asked / hidden), items by kind	Watch for sudden shifts

Evaluation criteria with measured values, justification for each target and example alert rules: `docs/monitoring.md`. Offline evaluation is Part B.

11. Privacy and responsible AI

Privacy. Alert messages may contain host names or user IDs. All data stays on the team's server (SQLite and a volume), behind the login. Messages are cut to 300 characters, and IDs and numbers are normalised before learning. Jobs and evaluations are pruned; marks keep only service and time. No data is sent to an external AI service; only paging summaries go to the team's own paging tools, in Live mode.

Responsible AI. Fairness between services: metric rules treat every service the same, and alerts are capped at a quarter of the budget, so services that log a lot cannot crowd out the rest. Every item carries a reason. Humans oversee through shadow mode, "ask a person" and marks. Hidden alerts stay visible. The system never acts on infrastructure. Queue data should not be used to rate engineers. Details: `docs/privacy_responsible_ai.md`.

12. Project scope and limitations

In scope (built): the causal budgeted queue (jumps, leaks, gaps, alerts); the alert classifier with a model registry, promotion gate and rollback; the web app, API and CLI; Alertmanager, Prometheus, PagerDuty, Slack and Opsgenie connectors; *Evaluate my data*; shadow mode and marks; security hardening; monitoring endpoints; Docker and per-environment settings; CI; the GAIA, NAB, SMD and RCAEval evaluations.

Out of scope (not built): automatic remediation; multi-tenant hosting and user roles; LLM summaries of incidents; CPU, disk or network fault detection inside microservices (measured only on SMD servers); a managed cloud deployment; browser UI tests; a live pilot with a real team (the next step).

Limitations (say these if asked):

5. **Every GAIA fault is the same 1 GB, 600 s memory program.** The robustness replay (§B3) tests other sizes and shapes. §B5 tests other domains (NAB, a shop, SMD servers) and the same memory injection on three other microservice applications (RCAEval), with small test sets and a chance line.

6. **The win over a plain Prometheus rate rule is modest** (+3.7 points for the detector, 8/8 weeks; +0.7 for the full queue, which ties it week by week). The larger gains are fewer false items, a real-time queue, and leak and gap coverage. Carry-over lifts both rules about equally.
7. **K = 20 is below the fault rate** (22 a day), so no method can exceed 84.9%.
8. **Learned thresholds do not transfer.** One learned on GAIA's 1 GB faults rejects 0.5 GB faults. Rules-only mode works without labels; the per-series confirm levels (§B4) are measured but not in the headline.
9. **The AI component adds no catches on GAIA.** With log alerts chosen by severity, or left out, the queue catches slightly more (71.0% and 71.2% vs 69.6% at K = 20) with fewer false items (§5, §B4). The classifier is kept for teams whose faults log and who mark items; that benefit is argued, not measured here. The promotion gate now checks this on every batch, and on GAIA it keeps the classifier out (§9, §B4).
10. The scripts in `app/experiments/legacy_alerts_only/` are alerts-only milestone experiments. Do not quote them as the system result.

Future work. RCAEval's CPU, disk, delay and packet-loss injections (only the memory injection is scored, in §B5); a pilot with one team in shadow mode, measuring precision from marks. R1 through R8 in `docs/threat_model.md` are closed and tested: the Prometheus fetch uses the address it checked, login locks out (behind a trusted proxy the client is read from the right of `X-Forwarded-For`, across every header line; a code review on 10 October found the leftmost, client-written value was used, which let a client dodge the lock, and that only the first header line was read; both are fixed with regression tests), paging is by item key, models load through skops, sign-in sets a session cookie, a viewer password cannot change settings, the audit log is also a file, and paging secrets stay out of `ops.json`.

13. Success criteria

Criterion	Measured	Met?
The main workflow (webhook → queue → page → mark → retrain) completes	Integration tests and the demo	Yes
At least 90% of predefined test cases pass	273 / 273	Yes
The system handles the agreed concurrent users	50 users, 0 errors	Yes
Average response time below the threshold	p50 0.21 s, p95 0.32 s (target 1 s)	Yes
AI output meets the predefined quality criteria	Detector beats the rate rule in 8/8 weeks with fewer false items; the full queue ties it; the alert classifier adds no catches on GAIA (§B4 ablation), and the promotion gate keeps it out there	Partly: the detector, yes; the full queue, a tie; the classifier, not shown on GAIA
No critical security vulnerabilities in testing	0 critical. R1 through R8 are closed and tested	Yes
Deployable from documented instructions	`docker compose up`, `docs/deployment.md`, CI Docker job	Yes
Another person can reproduce the environment	`uv sync`, lock files, `run_validate.py`	Yes
The agreed functional requirements are met	20 FR, each traced to tests (`docs/requirements.md`)	Yes

Part B — Results

B1. Test-period results (K items a day, 55 days, 1,223 faults)

Caught = a shown jump or leak on the fault's service overlaps the injection window, or a shown alert carries the fault's id. Gap items never count as catches.

Baselines are the rules a team would already have, all scored with the same causal queue:

System (K = 20 items a day)	Caught	Share of best possible	Items / day	False / day	Precision
Log severity + exact duplicates	0.0%	0%	0.8	0.8	0%
Static rule: memory > week-1 p99	57.2%	67%	16.0	3.0	81%
Rate rule: `delta(mem[5m]) > 0.5 GB`	68.9%	81%	17.4	2.1	88%
Jump rule, first come	67.8%	80%	17.5	2.4	86%
Jump rule + learned threshold (detector alone)	72.6%	86%	17.1	1.0	94%
Full queue (jumps + leaks + gaps + alerts)	69.6%	82%	17.7	1.8	90%
Old headline: day's jumps ranked by final peak (not real time)	*72.9%*	*86%*	*17.5*	*1.3*	*93%*

Best possible at K = 20 is 84.9%: GAIA injects 22 faults a day, more than 20 on 34 of 55 days.

At K = 5 / 10 / 50 the full queue catches 19.5% / 38.8% / 85.8% (rate rule: 19.3% / 38.6% / 86.0%). Its advantage is at the realistic budget K = 20, where slots are scarce, and in false items (1.0–1.8 a day vs 2.1).

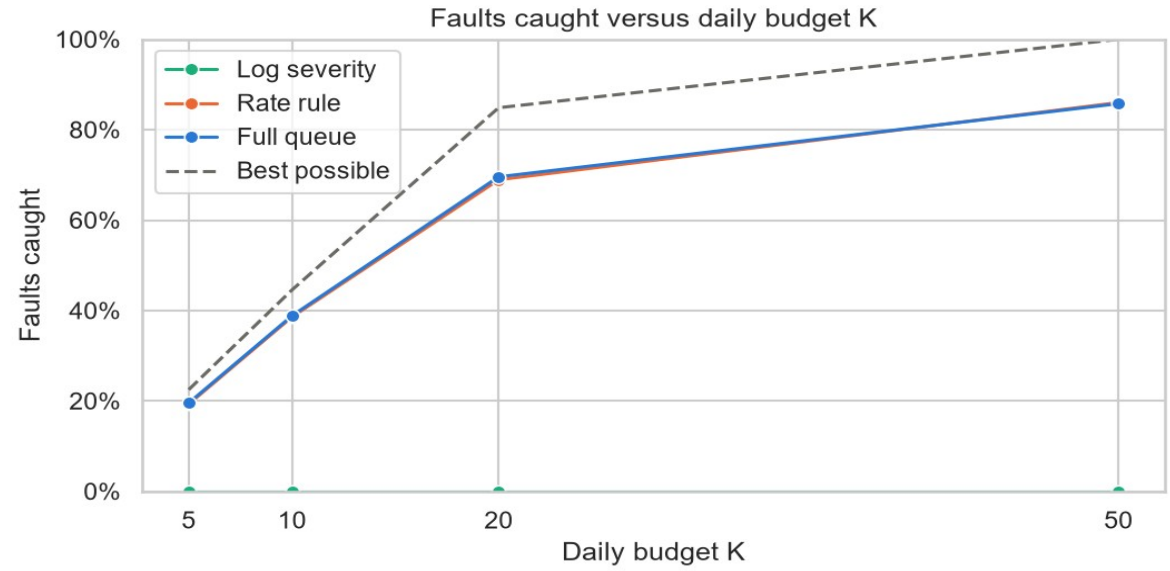
Removing the last look-ahead. A code review found five places where the queue still used information from later in the day, although the text said it did not. The per-day alert filters kept each day's *most severe* (or highest-scoring) K alerts, which needs the whole day; an overlapping jump that arrived *after* a leak was confirmed could still remove that leak; the robust score used the spread of the whole series; and an alert "rode on" a jump using the jump's final end. All five now use only what is known at the moment of the decision: alerts spend the budget first come, first served; a leak is dropped only by a jump already shown; the robust spread uses earlier points only; and an alert rides on a jump only while the jump's last hot point is within its gap (`Spike.open_at`). `tests/test_causality.py` checks each one. The honest cost is small: the full queue drops from 70.2% to 69.6% at K = 20 and shows 1.8 false items a day instead of 1.6. The detector alone is unchanged (72.6%), because none of the five touched it.

Why the full queue scores below the detector alone on GAIA. Every GAIA fault is a 1 GB memory fault, so leak items (0.2 a day), gap items (0.4 a day) and log alerts spend slots that jumps would have used. On a platform with leaks, outages, and CPU, disk or code faults, those are the parts that catch them (§B3). Both rows are reported; the detector comparison is the first claim. The full queue is the published configuration with the classifier on; the tool's promotion gate turns the classifier off on GAIA (§B4).

The old ranking needed the whole day in advance. Replacing it with a learned confirmation threshold keeps the score (72.6% vs 72.9%), cuts false items (1.0 vs 1.3 a day), and makes the queue usable live.

Time to confirm. Median **0.72 minutes** from fault start to the confirming point (p90 0.93). Metrics are sampled every 30 seconds.

Chance check. If each service's jumps are shifted by a random 1–6 hours, only **2.6%** of faults still overlap one.



Faults caught versus daily budget K

Where the misses come from

The full queue misses 372 of the 1,223 test faults at K = 20 (`results/miss_causes_gaia.csv`):

Cause	Faults	Share of misses
No metric data during the fault (monitoring outage)	166	45%
A qualifying jump existed, but the day's 20 slots were used	198	53%
Metric data, but no jump overlapped	4	1%
A jump overlapped, but below the confirmation threshold	4	1%

Only 2% of misses are the detector's fault. The rest are data outages (which the queue flags as gap items) and a budget smaller than the fault rate.

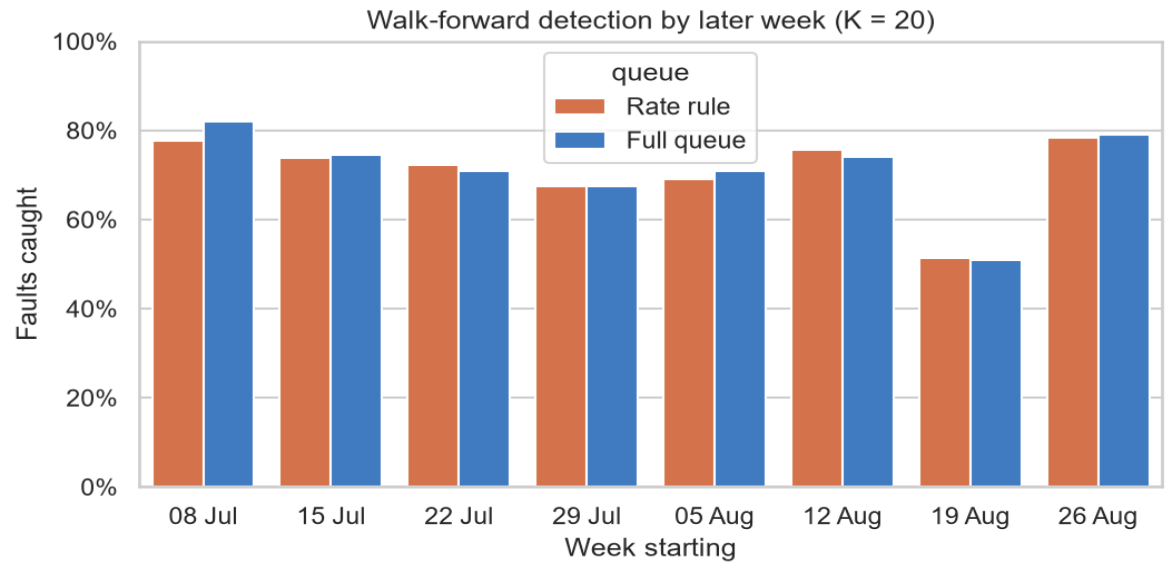
B2. Validation (walk-forward)

Before each later week, **everything learned is re-learned from the past only** — the classifier, its threshold, the jump ranker and the jump confirmation threshold (0.62–0.75 GB across folds). That week is then scored once. `run_walkforward.py`, K = 20.

Fold	Test week	Faults	With metric data	Rate rule	Detector alone	Full queue
1	8–15 Jul	161	100%	77.6%	84.5%	82.0%
2	15–22 Jul	134	91%	73.9%	81.3%	74.6%
3	22–29 Jul	162	94%	72.2%	77.2%	71.0%
4	29 Jul–5 Aug	163	82%	67.5%	69.3%	67.5%
5	5–12 Aug	162	84%	69.1%	71.6%	71.0%
6	12–19 Aug	139	92%	75.5%	79.1%	74.1%
7	19–26 Aug	173	61%	51.4%	52.0%	50.9%
8	26–31 Aug	129	91%	78.3%	83.7%	79.1%
Mean				70.7%	74.8%	71.3% (95% CI 64.5–76.7%)

- The **detector** (jump rule + learned threshold) beats the rate rule in **8/8** weeks (sign test p = 0.0039).
- The **full queue** does **not** beat the rate rule week by week: 4 wins, 3 losses, 1 tie (p = 0.5). Before the look-ahead fixes in §B1 it won 6 of 8; part of that margin came from choosing each day's alerts with hindsight. On GAIA its extra item types cost slots; see §B1.
- The comparison is against a real rule. The log-severity rule scores 0% every week, so a test against it would prove nothing.

Fold 7 is weak because of a **monitoring outage**: every service's metrics stop from 20 Aug 13:18 UTC for 2 days 12 hours, so only 61% of that week's faults have any metric data. The full queue shows this outage as a gap item within 10 minutes. No metric-based method could catch those faults.



Walk-forward detection by later week

B3. Robustness: does it only work because every fault is 1 GB?

`run_robustness.py` adds synthetic faults of other shapes to the **real** GAIA memory series (120 per shape, at least an hour from any real fault) and asks whether each detector fires, with no budget:

Injected shape	Jump rule 0.5 GB	+ learned threshold	Relative +30%	Rate rule	Jump + leak rule
Step +0.25 GB, 10 min	1%	0%	8%	1%	1%
Step +0.5 GB, 10 min	99%	0%	58%	99%	99%
Step +1 GB, 10 min (GAIA shape)	100%	98%	95%	100%	100%
Step +2 GB, 10 min	100%	100%	100%	100%	100%
Spike +1 GB, 1 min	98%	98%	92%	100%	98%
Ramp +1 GB over 30 min	1%	1%	6%	0%	99%
Leak +1 GB over 3 h	12%	3%	6%	8%	73%
Nothing injected (control)	6%	3%	6%	6%	6%

What this shows:

- **Faults smaller than the jump size are invisible** (0.25 GB). A team must set the size for its own services.
- **The learned threshold over-fits the benchmark**: trained on 1 GB faults, it rejects every 0.5 GB fault. The tool therefore uses **no** confirmation threshold unless it is trained on the team's own incidents.

- **Leaks and ramps defeat both the jump rule and the Prometheus rate rule.** The leak rule catches 99% of 30-minute ramps and 73% of 3-hour leaks, at a cost of 0.3 extra items a day on real GAIA data (none of them near a fault), and no extra hits on the control windows.

B4. Queue options

Option (K = 20)	Caught	Items / day	Walk-forward mean	Verdict
Full queue (reference)	69.6%	17.7	71.3%	—
Learned jump ranker instead of the rise threshold	69.7%	17.7	71.8%	No gain: the rise threshold already gives 94% precision, so there is little left to learn from service, time of day or nearby logs
Group metric items within 2 min on other services into one slot	70.7%	17.6	72.6%	Small gain; one item may then name two services
Confirm level per series (learned on week 1; off in the headline)	69.3%	17.6	70.5%	Mixed: the detector alone keeps 72.6% with fewer false items (0.8 vs 1.0 a day), but the levels reject real faults — 26 misses below threshold instead of 4, and 84.1% vs 85.8% at K = 50. The web app uses it; the published tables use the single gate
Carry over up to 10 unused slots to the next day	72.6%	18.6	74.8%	Largest gain; the average stays under K. But the same carry-over lifts the rate rule to 72.5% (walk-forward 74.3%), and the full queue then wins only 3 of 6 decided weeks (p = 0.66). Before the look-ahead fixes this row read 74.4% and 8/8 weeks; that margin did not survive

Ablation: what the alert classifier adds (`run_classifier_ablation.py`, `results/classifier_ablation_gaia.csv`). Same full queue, same week-1 training; only the choice of log alerts for their quarter of the slots changes. The classifier row reproduces the published full-queue numbers exactly.

Log alerts in the queue (K = 20)	Caught	Items / day	False / day	Precision
Chosen by the classifier (the full queue)	69.6%	17.7	1.8	90%
Chosen by log severity + exact duplicates	71.0%	17.6	1.4	92%
None (metric items only)	71.2%	17.4	1.2	93%

On GAIA the classifier does not earn its slots: on net, the alerts it escalates catch fewer faults than the metric items they push out of a full day. At K = 50, where slots are not scarce, all three catch 85.8%. This is why the detector alone is the first claim (§B1) and why the classifier is described as an option for teams whose faults log (§5).

Carry-over and grouping are off by default (`CARRY_OVER_MAX`, `GROUP_MINUTES`): they change what "20 a day" means for the on-call team, so it is their choice.

Why the severity row attaches no alerts. An alert rides on a jump (no slot of its own) only if it arrives while a shown metric item on *its own service* is still open (`build_queue`). The classifier row attaches 280 alerts that way. The severity row attaches none, and the data explains why (`results/classifier_ablation_gaia.json`). The 507 test alerts that carry a fault are all `warning` (471) or `info` (36); no test alert is `critical` or `high`. So the severity rule never escalates a fault's alert. It escalates 44 `error` lines, with 14 more cut by its own 20-a-day limit. All 44 fall on 4 days, none carries a fault, and they are nightly jobs reporting that a log upload failed. None arrives while a shown metric item on its service is open:

the nearest one on the same service is at least 199 minutes away (median 617). So 11 of them take a slot of their own, the rest find the day's alert share or budget used, and the row ends up close to "no log alerts".

What the shipped gate does with this. The tool trains the classifier on the first 7 days of a batch and tests it on the rest (§9). A model must cost less than showing no log alerts on the held-out alerts, and the whole held-out queue with its alerts must cost less than the same queue without log alerts. The first model is not promoted just because no model is active. On GAIA the classifier passes the first test: it finds 339 of the 507 fault-bearing held-out alerts. It fails the second: with its alerts the queue catches 848 of 1,223 held-out faults with 95 false items, against 872 with 58 false items without log alerts. So it is registered as a candidate, and GAIA is scored by rules only (log alerts by severity, as in the second row). The gate's queue is the tool's, not this table's (it uses the tool's own confirm settings), so its counts differ a little from the rows above; the direction is the same. `run_classifier_ablation.py` records the decision in the same JSON file.

B5. Beyond microservices: any metric, any team

The engine is not tied to GAIA. `src/generic.py` takes any **metrics** table (time, value, optional series; long or wide; column names detected), optional **events** (alerts, tickets, notifications) and optional **past problems** (start, end, series). For each series it tunes on the first part and scores the rest once, and every system — static threshold, rate rule, a unit-free robust jump rule, the user's own alerts — is tuned with the same cost and scored through the same causal queue. The detector with the lowest training cost becomes the user's queue. A **chance line** (the same items at random times) is reported so wide labels cannot flatter the result.

This runs from the web app (**Evaluate my data**: upload or pick a dataset → read the report → *Use these settings for my live queue* → *Show this data in the queue*) and from the CLI (`evaluate`, `tune`). The guide for end users is `USE_YOUR_DATA.md`.

Results on data that is not microservices (tuned on the first half of each series, K = 2 items a day; `run_nab.py`, `results/nab_eval.csv`):

Dataset	Domain	Test problems	Queue caught	Chance	Items / day
Online shop (synthetic; K = 5, 30% tuning)	e-commerce orders + notifications	11	91%	1%	1.3
NAB realAWSCloudwatch	cloud servers	17	71%	58%	0.3
NAB realAdExchange	online advertising	6	83%	58%	1.4
NAB realKnownCause	machine temperature, taxi demand, CPU	13	38% (static rule 54%)	15%	0.04
NAB realTraffic	road traffic sensors	11	36% (rate rule 45%)	12%	0.3
NAB realTweets	Twitter mention volume	11	36%	36%	1.7

On the shop data the monitoring view catches 10 of 11 problems where the notification stream by severity catches 3. On NAB the queue is clearly above chance on traffic and machine data, modestly on cloud servers and ads (NAB's anomaly windows are wide, so chance is high), and **no better than chance on Twitter volume**. The test sets are small (6–17 problems), and on one dataset the detector chosen on the tuning part was not the best on the test part. These are exactly the facts a new user needs before relying on it, which is why the app shows them.

Other fault types: the Server Machine Dataset (SMD). GAIA has one fault type. SMD (OmniAnomaly, KDD 2019) has 28 real servers, five weeks of one-minute samples, 38 metrics each — CPU, memory, disk, network, TCP — and 196 labelled test incidents, each with the metrics that moved. Each machine is one dataset: tuned on the

first half of every series, scored on the second half, K items a day per machine (`run_smd.py`, `results/smd_eval.csv`):

Budget K per machine	Incidents caught (196)	Metric movements shown (1,536)	Items / day	False / day	Chance (per movement)
5	60 (30.6%)	10.1%	4.2	3.5	12.4%
10	83 (42.3%)	19.3%	7.9	6.5	12.4%
20	108 (55.1%)	33.9%	13.8	11.0	12.4%

An incident counts once and is caught if any metric that moved was shown while it lasted; the movement column counts every flagged metric on its own. Half of the incidents are found at 20 items a day per machine on real mixed-metric faults, well above the 12.4% chance line, but with low precision: SMD labels only some of the 38 metrics per incident, so a correlated movement on an unlabelled metric counts as false. Where a series had three or more tuning incidents the jump rule learned its own size (Section 4 of `USE_YOUR_DATA.md`). This is a harder benchmark than GAIA and it is reported as it is.

Same memory fault, three applications (RCAEval RE1). RE1 injects one memory fault, five times on each of five services, into Online Boutique, Sock Shop and Train Ticket (75 recordings; RCAEval). Each recording is scored on its own. The first 12 recordings of an application tune the queue; the later 13 are scored once. Those later recordings are then laid end to end, about one day, and the daily cap is applied. Thresholds come from that application's own training recordings (`run_rcaeval.py`, `results/rcaeval_mem.csv`).

The labelled service's memory series does move: median rise 0.51 GB, 0.30 GB and 0.33 GB. With the cap off, the jump rule overlaps 100%, 92.3% and 61.5% of the later injections, each above that rule's chance line (28.5%, 47.1%, 40.9%).

Application	Memory series	Median rise	Jump episodes on the test recordings	Jump, no cap	Chance	Queue at K = 5
Online Boutique	12	0.51 GB	744	100%	28.5%	0%
Sock Shop	15	0.30 GB	235	92.3%	47.1%	0%
Train Ticket	64	0.33 GB	1,512	61.5%	40.9%	0%

The queue that would page someone catches 0 of 13 faults on every application at K = 2, 5 and 10. On the training recordings the jump rule caught all 12 faults and also raised 679, 220 and 1,323 other episodes. At the usual cost (a miss counts 10, a false episode counts 1) that is more expensive than a rate rule which raises nothing, so the tuner keeps the rate rule. Under the same cap the jump rule still misses: the test day contains hundreds of episodes and five slots, and the slots go to episodes that miss the injection. The fault is visible on all three applications. This budget and this cost leave it off the page.

B6. What "success" means on this data

`run_stats.py` splits the 1,223 test faults by the signal they left:

Signal	Count	Share	What can catch it
Jump, no log	868	71.0%	Metric rule only
Log and jump	185	15.1%	Either path
Log, no jump	34	2.8%	Alerts only
No log, no metric data (monitoring gap)	132	10.8%	Nothing — shown as a gap item
No log, data but no jump	4	0.3%	Nothing

- **Where metric data exists, the jump rule overlaps 99.6% of faults** (1,053 / 1,057).
- 170 test faults start inside a monitoring gap. At K = 20 the gap items shown cover 159 of them (all 170 at K = 50).
- **False jumps are not mislabelled faults.** 24.6% of the 256 false jumps start within 10 minutes of a fault on another service; random times do so 27.2% of the time. There is no sign that GAIA logged faults under the wrong service.
- A log-only classifier is capped near **17.9%** (the faults that leave any log). That is why the early 95% target was wrong.

Early target	Measured	Verdict
95% detection at K=20	69.6–72.6% (ceiling 84.9%)	Not possible at K=20: more faults per day than slots. 82–86% of the ceiling
Beat the usual rule	Detector +3.7 points over a rate rule, 8/8 weeks; full queue +0.7 points, 4 of 8 weeks; log severity 0%	Met by the detector, against a fair rule; the full queue ties it
Fewer false items	1.0–1.8 vs 2.1 a day (rate rule)	Met
Works on every service	Jump overlap 81–91% on all ten	Met
Walk-forward not lucky	Means 71.3–74.8%, CI 64.5–76.7%	Met
Real-time	Confirmed in 0.72 min; no look-ahead, checked by `tests/test_causality.py`	Met
Works beyond 1 GB faults	0.5–2 GB steps and spikes ≥ 98%; leaks 73–99% with the leak rule; 0.25 GB not	Partly — stated

Per service (test period):

Service	Faults	With a log	With a jump
dbservice1	129	0.8%	91.5%
dbservice2	148	2.0%	81.1%
logservice1	126	2.4%	87.3%
logservice2	108	1.8%	87.0%
mobservice1	124	84.7%	83.9%
mobservice2	116	87.1%	87.1%
redisservice1	111	2.7%	82.0%
redisservice2	114	0.0%	86.0%
webservice1	122	0.8%	85.2%
webservice2	125	0.0%	90.4%

Part C — Reproduce and deliverables

C1. How to reproduce

```
python tools/fetch_gaia.py        # needs internet, ~1.8 GB download; checks SHA-256
cd app
$env:PYTHONPATH = (Get-Location).Path
python -m pytest -q
python run_memory.py --metrics data/gaia_metrics --gaia-run data/gaia_raw/run/run --sweep
python run_queue_eval.py
python run_classifier_ablation.py
python run_walkforward.py
```

```
python run_stats.py
python run_robustness.py
python ../tools/fetch_nab.py      # needs internet, ~15 MB
python run_nab.py
python ../tools/fetch_smd.py      # needs internet, ~240 MB
python run_smd.py                # 28 machines, about 15 min; resumes if interrupted
python ../tools/fetch_rcaeval_mem.py   # needs internet; RE1 memory cases only
python run_rcaeval.py            # three applications, a few minutes
python make_figures.py
python run_validate.py
```

Needs the GAIA files at `app/data/gaia_raw/run/run` and `app/data/gaia_metrics` (not in git; they are large). `python tools/fetch_gaia.py` (from the repository root) downloads them from the GAIA MicroSS dataset (https://github.com/CloudWise-OpenSource/GAIA-DataSet, `MicroSS/run/run.zip` and the split archive `MicroSS/metric/metric_split.*`, about 1.8 GB), extracts the two run tables and the 37 `docker_memory_usage_total` series into that layout, and checks each file's SHA-256 against the copy used for these results (`--check` verifies files already in place). Without GAIA: `python examples/make_example.py`, then `python -m product.cli score --alerts examples/alerts.csv --metrics examples/metrics.csv`.

A faster check without GAIA: `cd app && uv sync --group dev && uv run pytest -q && uv run python run_validate.py` (the result files are in git). Every experiment run is logged in `app/results/experiments.jsonl` (`docs/mlops.md`).

C1a. Use of AI tools

AI coding assistants were used while building this project: for code review, refactoring, test writing and drafting documentation (visible in the repository history as the `claude/*` branches merged through pull requests #2 to #5, and in `AGENTS.md`). The runtime system uses no generative AI or external AI service. Every number in this report comes from the scripts in `app/` and is checked against `app/results/` by `run_validate.py`; the test suite, lint, type checks and the dependency audit run in CI. Design decisions, the evaluation method and the claims made here are the author's responsibility.

C2. Deliverables

Item	Where
Design	`design_specification.md`
Requirements and traceability	`docs/requirements.md`
Architecture diagrams	`docs/architecture.md`
Design decisions	`docs/adr/`
Threat model	`docs/threat_model.md`
Test plan and CI gates	`docs/test_plan.md`, `.github/workflows/ci.yml`
Deployment, environments, AI governance	`docs/deployment.md`, `app/deploy/*.env`
MLOps: versioning, updates, experiment tracking	`docs/mlops.md`, `app/results/experiments.jsonl`
Monitoring and evaluation	`docs/monitoring.md`, `/metrics`, `/monitoring`
Privacy and responsible AI	`docs/privacy_responsible_ai.md`
Performance results	`app/perf/load_test.py`, `app/results/perf/`
Retrospective	`docs/retrospective.md`
Change history	`CHANGELOG.md`
Plan / roadmap	`project_plan.md`, `ROADMAP.md`
Validation record	`VALIDATION.md`

Item	Where
Review of the earlier version	CODE_REVIEW.md
This report	FINAL_REPORT.md
Code + tests	app/
Result tables	app/results/
Figures	figures/detection_vs_k.svg, figures/walkforward.svg
Tool	Web app docker compose up (http://127.0.0.1:8080/, /evaluate for your own data), CLI python -m product.cli
End-user guide	USE_YOUR_DATA.md

Checked numbers

python run_validate.py (in app/) recomputes every value below from app/results/ and fails if one no longer rounds to what is printed here, if a key is missing, or if a sentence it guards (for example "the detector wins 8/8 weeks") stops being true. Rates are shares of test faults; false is false items a day.

Key	Value	Where in this report
queue.K20.logs_severity	0.0%	§B1 table
queue.K20.static_p99	57.2%	§B1 table
queue.K20.delta_5m	68.9%	§B1 table
queue.K20.jumps_online	67.8%	§B1 table
queue.K20.jumps_tuned	72.6%	§B1 table
queue.K20.proposed	69.6%	§B1 table
queue.K20.jumps_hindsight	72.9%	§B1 table
queue.K50.proposed	85.8%	§B1 text
false.K20.delta_5m	2.1	§B1 table
false.K20.jumps_tuned	1.0	§B1 table
false.K20.proposed	1.8	§B1 table
ceiling.K20	84.9%	§B1
misses.K20.no_metric_data	166	§B1 misses
misses.K20.budget_full	198	§B1 misses
misses.K20.no_jump	4	§B1 misses
misses.K20.below_threshold	4	§B1 misses
walkforward.proposed.mean	71.3%	§B2
walkforward.proposed.ci_low	64.5%	§B2
walkforward.proposed.ci_high	76.7%	§B2
walkforward.proposed.wins	4	§B2
walkforward.jumps_tuned.mean	74.8%	§B2
walkforward.jumps_tuned.wins	8	§B2
walkforward.delta_5m.mean	70.7%	§B2
robustness.step_1g.jump_0.5g	100%	§B3
robustness.step_0.25g.jump_0.5g	1%	§B3

Key	Value	Where in this report
robustness.ramp_1g.jump+leak_3h	99%	§B3
robustness.step_0.5g.jump_0.5g+threshold	0%	§B3
ablation.K20.classifier	69.6%	§5, §B4
ablation.K20.severity	71.0%	§5, §B4
ablation.K20.no_alerts	71.2%	§5, §B4
ablation.false.K20.classifier	1.8	§5, §B4
ablation.false.K20.severity	1.4	§5, §B4
ablation.false.K20.no_alerts	1.2	§5, §B4
queue.K20.p3_ranker	69.7%	§B4
queue.K20.p3_grouped	70.7%	§B4
queue.K20.p3_per_series	69.3%	§B4
ablation.attached.K20.classifier	280	§B4
severity.K20.fault_alerts	507	§B4
severity.K20.fault_alerts_warning	471	§B4
severity.K20.fault_alerts_info	36	§B4
severity.K20.escalated	44	§B4
severity.K20.escalated_with_fault	0	§B4
severity.K20.escalated_days	4	§B4
severity.K20.cut_by_daily_budget	14	§B4
severity.K20.escalated_while_shown_jump_open	0	§B4
severity.K20.minutes_to_shown_metric_min	199	§B4
severity.K20.minutes_to_shown_metric_median	617	§B4
severity.K20.alerts_shown	11	§B4
gate.held_out.fault_alerts_found	339	§B4
gate.queue.faults	1223	§5, §B4
gate.queue.caught_with_model	848	§5, §B4
gate.queue.caught_no_log_alerts	872	§5, §B4
gate.queue.false_with_model	95	§B4
gate.queue.false_no_log_alerts	58	§B4
queue.K50.p3_per_series	84.1%	§B4
false.K20.p3_per_series_jumps	0.8	§B4
queue.K20.p3_carryover	72.6%	§B4
queue.K20.delta_5m_carryover	72.5%	§B4
walkforward.carryover.mean	74.8%	§B4
walkforward.delta_carryover.mean	74.3%	§B4
nab.realAWSCloudwatch	71%	§B5
nab.realAdExchange	83%	§B5
nab.realKnownCause	38%	§B5
nab.realTraffic	36%	§B5
nab.realTweets	36%	§B5

Key	Value	Where in this report
smd.K5.incidents	30.6%	§B5
smd.K10.incidents	42.3%	§B5
smd.K20.incidents	55.1%	§B5
smd.K5.movements	10.1%	§B5
smd.K10.movements	19.3%	§B5
smd.K20.movements	33.9%	§B5
rcaeval.online-boutique.rise_gb	0.51	§B5
rcaeval.online-boutique.jump_spikes	744	§B5
rcaeval.online-boutique.jump_open	100%	§B5
rcaeval.online-boutique.jump_chance	28.5%	§B5
rcaeval.online-boutique.jump_train_false	679	§B5
rcaeval.online-boutique.K5	0%	§B5
rcaeval.sock-shop.rise_gb	0.30	§B5
rcaeval.sock-shop.jump_spikes	235	§B5
rcaeval.sock-shop.jump_open	92.3%	§B5
rcaeval.sock-shop.jump_chance	47.1%	§B5
rcaeval.sock-shop.jump_train_false	220	§B5
rcaeval.sock-shop.K5	0%	§B5
rcaeval.train-ticket.rise_gb	0.33	§B5
rcaeval.train-ticket.jump_spikes	1512	§B5
rcaeval.train-ticket.jump_open	61.5%	§B5
rcaeval.train-ticket.jump_chance	40.9%	§B5
rcaeval.train-ticket.jump_train_false	1323	§B5
rcaeval.train-ticket.K5	0%	§B5
stats.jump_rate_with_data	99.6%	§B6
stats.chance_overlap	2.6%	§B1
stats.confirm_minutes_median	0.72	§B1
stats.false_jumps_near_fault	24.6%	§B6